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Professor Summerfield
Human Information Processing Lab
Department of Experimental Psychology
Oxford University

Dear Professor Summerfield,

I am writing to express my interest in applying for a PhD position in the Human Information Processing Lab. My research interests lie at the intersection of game theory, cooperation, complex networks, and artificial intelligence. In particular, I am interested in understanding the conditions under which cooperation can emerge, persist, or be restored in strategic interactions, and how AI-based mechanisms may be used to promote cooperative and socially beneficial behaviour.

I am particularly drawn to your lab's research theme, "Using AI to promote cooperation". I find this direction compelling because it addresses collective-action problems such as sustainability, social justice, and democratic coordination through the design of AI systems that can assist human groups fairly and beneficially. I first became interested in this line of work around June 2025, when I read *Deep reinforcement learning can promote sustainable human behaviour in a common-pool resource problem*. I was especially inspired by the idea of treating a reinforcement learning agent as a social planner that can improve collective outcomes while also considering fairness and sustainable resource use in interactions with human participants. This led me to further study related work associated with this research programme, including *Scaffolding cooperation in human groups with deep reinforcement learning*, *Deep mechanism design: learning social and economic policies for human benefit*, and recent work on AI-assisted democratic deliberation. These studies strongly resonate with my own interest in using formal models and AI tools to understand and promote cooperation.

This line of research has directly influenced my recent work. In my paper *Recovering Cooperation in Spatial Prisoner's Dilemma via a Forced Loner Mechanism*, published in *Physica A: Statistical Mechanics and its Applications*, I studied how introducing a protective intervention can affect cooperation, fairness, and collective outcomes in spatial social dilemmas. In another recent paper, *Exploring Cooperation Mechanisms via Reinforcement Learning in Network Common-Pool Resource Games*, available on *arXiv*, I extended the common-pool resource game to a networked structure and used reinforcement learning to search for and interpret resource allocation mechanisms suitable for structured populations. Although these projects were based mainly on modelling and simulation rather than human-participant experiments, they strengthened my interest in moving toward research that combines AI agents, behavioural experiments, and mechanism design for cooperation.

I believe my background would allow me to contribute to research in your group. Methodologically, I have experience with numerical simulation, reinforcement learning, network-based modelling, and the analysis of strategic interactions. Theoretically, I have developed a strong foundation in game theory, evolutionary dynamics, complex networks, and mathematical mod-

elling through my master's research. My work on tripartite intervention-conflict games and frequency-adjusted Q-learning has also given me experience in connecting game-theoretic models with adaptive learning dynamics. More broadly, I am interested in how classical models of social dilemmas can be extended to human–AI interaction settings, and how AI systems may help discover mechanisms that improve collective welfare.

For my PhD, I hope to study how AI systems can reshape cooperation in human and human–AI groups. More specifically, I am interested in how new behavioural patterns, cooperative norms, and collective outcomes emerge when humans interact with AI agents in social dilemma games. I hope to formalise these interactions using game-theoretic and complex-network models, and to examine how AI agents may act as mediators, social planners, or adaptive intervention mechanisms that help groups coordinate behaviour, resolve conflicts, and achieve fairer collective outcomes. I would be very grateful for the opportunity to explore whether my background and research interests aligns closely with the Human Information Processing Lab and its work on using AI to promote cooperation.

Yours sincerely,

Yihang Qin